GCSE Mathematics

Predicted Paper 3 — Higher Tier

Set E

Pearson Edexcel style (1MA1 / 3H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

• Calculator • Ruler graduated in centimetres and millimetres
• Protractor • Pair of compasses • Pen, HB pencil, eraser

Instructions

• Answer **all** 22 questions in the spaces provided.
• You must show all your working.
• Diagrams are NOT accurately drawn unless stated otherwise.
• Calculators may be used.
• If your calculator does not have a π button, take π to be 3.142.

Advice

• Read each question carefully before you start.
• The marks for each question are shown in brackets — use them as a time guide.
• Try to answer every question. Check your answers if you have time at the end.

Answer ALL questions. Write your answers in the spaces provided.

1. Write 168 as a product of its prime factors.

(2 marks)

2. It takes 6 machines 14 hours to complete a job. Assuming all machines work at the same rate, how long would it take 8 machines to complete the same job?

(3 marks)

3. Solve $7 - 2x < 15$.

(2 marks)

4. Write $2.4 \times 10^3 \times 5 \times 10^4$ in standard form.

(2 marks)

5. The two-way table shows the results of 100 driving test candidates.

	Pass	Fail	Total
Theory	36	14	50
Practical	42	8	50
Total	78	22	100

(a) Find the probability that a randomly chosen candidate passed.

(1 marks)

(b) Given that a candidate took the practical test, find the probability that they failed.

(2 marks)

(c) Are the events 'passing the test' and 'taking the theory test' independent? Justify your answer using a probability calculation.

(3 marks)

6. Sarah's salary is £32,500. She receives a 4% pay rise. Calculate her new salary.

(2 marks)

7. Solve $x^2 - 10x + 21 = 0$ by factorising.

(3 marks)

8. y is directly proportional to x^2. When x = 4, y = 80.

(a) Find a formula relating y and x.

(2 marks)

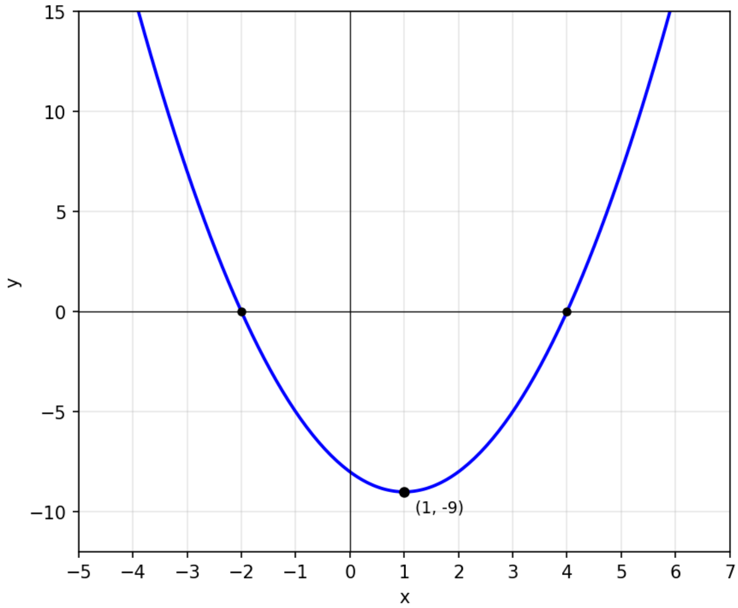
(b) Find the value of y when x = 7.

(2 marks)

(c) Find the positive value of x when y = 245.

(2 marks)

9. The graph shows the curve $y = x^2 - 2x - 8$. The curve crosses the x-axis at –2 and 4, and has turning point (1, –9).



(a) Use the graph to write down the values of x for which $x^2 - 2x - 8 < 0$.

(2 marks)

(b) Solve $x^2 - 2x - 8 = 7$. Show your method.

(3 marks)

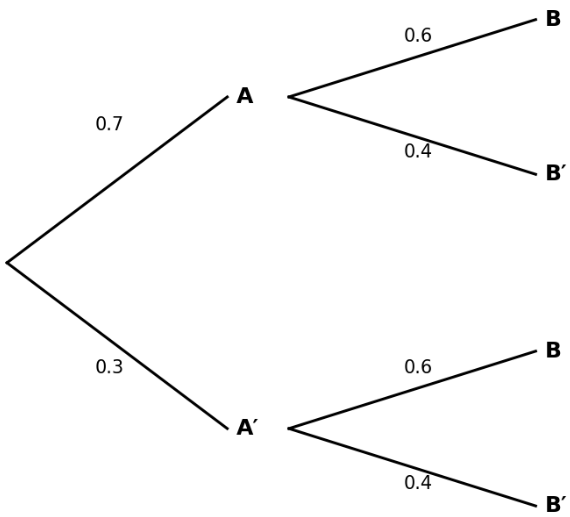
10. Rationalise the denominator of 12 / √6. Give your answer in its simplest form.

(2 marks)

11. A car travels from City A to City B at an average speed of 75 km/h. On the return journey from B to A, the average speed is 60 km/h. The total journey time (out and back) is 4.5 hours. Calculate the distance from City A to City B.

(4 marks)

12. A weather forecast gives the probability of rain on any given day as 0.3. If it rains, the probability that Alex cycles to work is 0.4; if it doesn't rain, the probability that he cycles is 0.6.



(a) Use the tree diagram (or otherwise) to work out the probability that Alex cycles on a given day.

(3 marks)

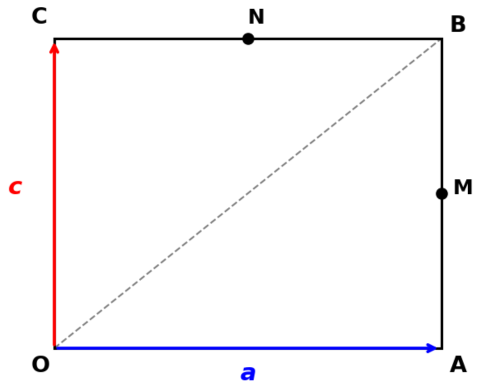
(b) Given that Alex cycled on a particular day, find the probability that it rained.

(3 marks)

13. Simplify fully $(4x^2 – 1) / (2x^2 + 5x – 3)$.

(4 marks)

14. OABC is a rectangle. **OA** = **a** and **OC** = **c**. M is the midpoint of AB and N is the midpoint of BC.



(a) Express the vector **MN** in terms of **a** and **c**. Simplify your answer.

(4 marks)

(b) What does your answer to (a) tell you about the relationship between MN and OA + OC? (Hint: compare directions and ratios.)

(2 marks)

15. Express $4 / (3 - \sqrt{5})$ in the form $a + b\sqrt{5}$ where a and b are rational numbers.

(3 marks)

16. Line L passes through A(2, 3) and B(8, –1).

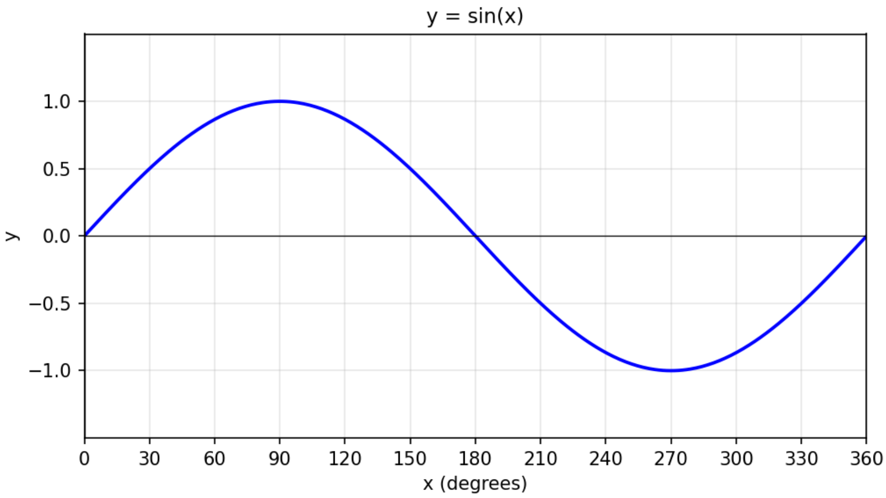
(a) Work out the gradient of line L.

(2 marks)

(b) Find the equation of the line that is perpendicular to L and passes through the midpoint of AB. Give your answer in the form $y = mx + c$.

(4 marks)

17. The graph shows $y = \sin(x)$ for $0° \leq x \leq 360°$.



(a) Use the graph (or otherwise) to write down all solutions of $\sin(x) = 0.5$ in the range $0° \leq x \leq 360°$.

(2 marks)

(b) Find all solutions of $\sin(x) = -0.5$ in the range $0° \leq x \leq 360°$.

(2 marks)

18. The value of an investment increases by 5% in the first year, by 8% in the second year, and by 6% in the third year.

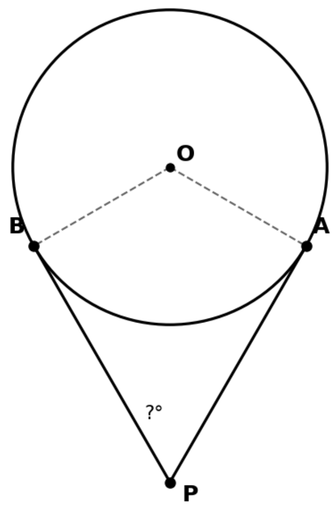
(a) Calculate the overall percentage increase over the 3 years. Give your answer to 2 decimal places.

(4 marks)

(b) If the original investment was £4,000, work out its value at the end of year 3.

(2 marks)

19. PA and PB are tangents to a circle with centre O, drawn from external point P. The radius of the circle is 2.5 cm and OP = 5 cm.



(a) Work out the size of angle OPA.

(2 marks)

(b) Hence write down the size of angle APB.

(1 marks)

(c) Work out the length of tangent PA. Leave your answer in surd form.

(3 marks)

20. A rectangle has length (x + 3) cm and width (x – 2) cm. The area of the rectangle is 50 cm^2.

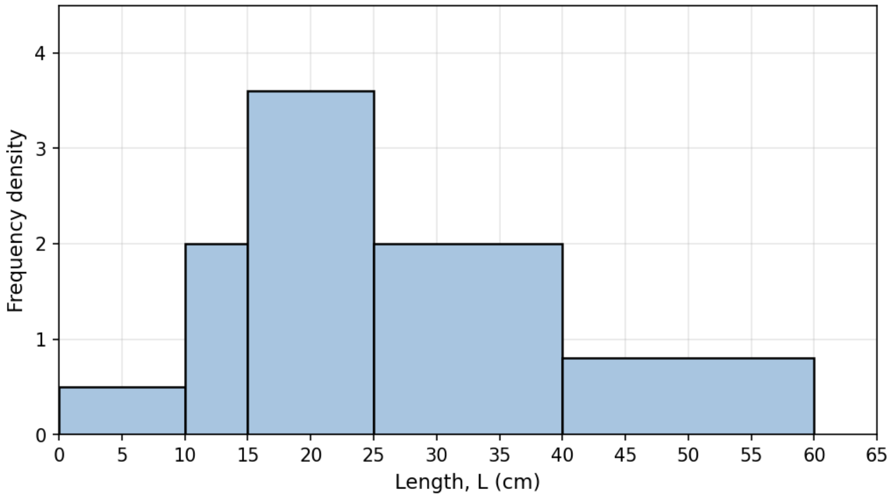
(a) Show that $x^2 + x - 56 = 0$.

(2 marks)

(b) Solve the equation to find the length and width of the rectangle.

(3 marks)

21. The histogram shows the lengths of 100 fish in a sample.



(a) Estimate the number of fish with length between 20 cm and 30 cm.

(2 marks)

(b) What percentage of fish are shorter than 15 cm?

(2 marks)

22. The first three terms of a geometric sequence are 5, 15, 45, ...

(a) Write down the common ratio.

(1 marks)

(b) Find an expression for the nth term of this sequence.

(2 marks)

(c) The kth term of the sequence is 32,805. Find the value of k.

(3 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 3 Set E

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	$2^3 \times 3 \times 7$	M1: factor tree or successive division ($168 = 2 \times 84 = 4 \times 42 = 8 \times 21 = 2^3 \times 21$). A1: $2^3 \times 3 \times 7$. **[2]**
2	10.5 hours (10 hours 30 min)	M1: total machine-hours = 6 × 14 = 84. M1: 84 / 8. A1: 10.5 hours. **[3]**
3	$x > -4$	M1: $7 - 2x < 15 \Rightarrow -2x < 8$. A1: $x > -4$ (note inequality reverses when dividing by negative). **[2]**
4	1.2×10^8	M1: $2.4 \times 5 = 12$; $10^3 \times 10^4 = 10^7$. A1: $12 \times 10^7 = 1.2 \times 10^8$. **[2]**
5(a)	78/100 = 39/50	B1: 78/100 or equivalent. **[1]**
5(b)	8/50 = 4/25	M1: identifies 8 fails out of 50 practical candidates. A1: 4/25 (or 8/50). **[2]**
5(c)	Not independent	M1: P(pass) = 0.78; P(theory) = 0.50; P(pass) × P(theory) = 0.39. M1: P(pass ∩ theory) = 36/100 = 0.36. A1: 0.36 ≠ 0.39, so the events are NOT independent. **[3]**
6	£33,800	M1: 32500 × 1.04. A1: 33,800. **[2]**
7	$x = 3$ or $x = 7$	M1: factorises $(x - 3)(x - 7)$. A1: $x = 3$. A1: $x = 7$. **[3]**
8(a)	$y = 5x^2$	M1: $y = kx^2$. $80 = 16k \Rightarrow k = 5$. A1: $y = 5x^2$. **[2]**
8(b)	$y = 245$	M1: $y = 5 \times 49$. A1: 245. **[2]**
8(c)	$x = 7$	M1: $245 = 5x^2 \Rightarrow x^2 = 49$. A1: $x = 7$. **[2]**
9(a)	$-2 < x < 4$	M1: identifies x-values where curve is below x-axis (between the roots). A1: $-2 < x < 4$. **[2]**
9(b)	$x = 5$ or $x = -3$	M1: rearranges: $x^2 - 2x - 15 = 0$. M1: factorises $(x - 5)(x + 3)$. A1: $x = 5$ or $x = -3$. **[3]**
10	$2\sqrt{6}$	M1: $12/\sqrt{6} \times \sqrt{6}/\sqrt{6} = 12\sqrt{6} / 6$. A1: $2\sqrt{6}$. **[2]**
11	150 km	M1: let d = distance one way. Time out = d/75; time back = d/60. M1: d/75 + d/60 = 4.5. M1: common denominator 300: 4d/300 + 5d/300 = 4.5 ⇒ 9d/300 = 4.5 ⇒ 9d = 1350. A1: d = 150 km. **[4]**
12(a)	0.54	M1: P(rain ∩ cycle) = 0.3 × 0.4 = 0.12. M1: P(no rain ∩ cycle) = 0.7 × 0.6 = 0.42. A1: total P(cycle) = 0.12 + 0.42 = 0.54. **[3]**
12(b)	0.222 (2/9)	M1: conditional P(rain | cycle) = P(rain ∩ cycle) / P(cycle). M1: = 0.12 / 0.54. A1: = 2/9 ≈ 0.222 (3 s.f.). **[3]**
13	$(2x + 1) / (x + 3)$	M1: numerator (difference of squares): $4x^2 - 1 = (2x - 1)(2x + 1)$. M1: denominator: factors of $2 \times -3 = -6$ summing to 5 are +6 and −1, giving $2x^2 + 6x - x - 3 = 2x(x + 3) - (x + 3) = (2x - 1)(x + 3)$. M1: cancels common factor $(2x - 1)$. A1: $(2x + 1)/(x + 3)$. **[4]**
14(a)	**MN** = −(1/2)**a** + (1/2)**c**	M1: B = **a** + **c** (diagonal of rectangle from O). M1: **OM** = (1/2)(**OA** + **OB**) = (1/2)(**a** + **a** + **c**) = **a** + (1/2)**c**. M1: **ON** = (1/2)(**OB** + **OC**) = (1/2)(**a** + **c** + **c**) = (1/2)**a** + **c**. A1: **MN** = **ON** − **OM** = −(1/2)**a** + (1/2)**c**. **[4]**
14(b)	MN is parallel to CA	M1: **MN** = (1/2)(−**a** + **c**) = −(1/2)(**a** − **c**). Also **CA** = **a** − **c**. A1: **MN** = −(1/2)**CA**, so MN is parallel to the diagonal CA (and half its length). **[2]**
15	$3 + \sqrt{5}$	M1: multiplies top and bottom by $(3 + \sqrt{5})$: $4(3 + \sqrt{5}) / (9 - 5)$. M1: $= (12 + 4\sqrt{5}) / 4$. A1: $3 + \sqrt{5}$, so a = 3, b = 1. **[3]**
16(a)	Gradient = −2/3	M1: gradient = (−1 − 3)/(8 − 2). A1: −4/6 = −2/3. **[2]**

Q	Answer	Mark scheme
16(b)	$y = (3/2)x - 6.5$	M1: midpoint = (5, 1). M1: perpendicular gradient = 3/2. M1: $y - 1 = (3/2)(x - 5)$. A1: $y = (3/2)x - 15/2 + 1 = (3/2)x - 13/2$ (or $1.5x - 6.5$). **[4]**
17(a)	$x = 30°$ or $x = 150°$	M1: identifies $\sin^{-1}(0.5) = 30°$. A1: lists both solutions in range. **[2]**
17(b)	$x = 210°$ or $x = 330°$	M1: notes sin is negative in 3rd and 4th quadrants; reference angle 30°. A1: 210° (180 + 30) and 330° (360 – 30). **[2]**
18(a)	20.20% (2 d.p.)	M1: multipliers $1.05 \times 1.08 \times 1.06$. M1: = 1.2020 (approx). M1: percentage increase = $(1.2020 - 1) \times 100$. A1: 20.20% (2 d.p.). **[4]**
18(b)	£4,808.16	M1: $4000 \times 1.05 \times 1.08 \times 1.06$. A1: 4808.16 (to nearest penny). **[2]**
19(a)	30°	M1: in right triangle OAP (right angle at A): sin(angle OPA) = OA/OP = 2.5/5 = 0.5. A1: angle OPA = $\sin^{-1}(0.5) = 30°$. **[2]**
19(b)	60°	B1: by symmetry, angle APB = 2 × angle OPA = 2 × 30 = 60°. **[1]**
19(c)	$(5\sqrt{3})/2$ cm	M1: Pythagoras: $PA^2 = OP^2 - OA^2 = 25 - 6.25 = 18.75$. M1: $PA = \sqrt{18.75} = \sqrt{(75/4)} = (\sqrt{75})/2$. A1: $(5\sqrt{3})/2$ (accept 4.33 cm or equivalent surd form). **[3]**
20(a)	Shown	M1: $(x + 3)(x - 2) = 50$. M1: $x^2 + x - 6 = 50$. A1: $x^2 + x - 56 = 0$. **[2]**
20(b)	length = 10 cm, width = 5 cm	M1: factorises $(x + 8)(x - 7) = 0$. M1: $x = 7$ (reject –8 as lengths must be positive). A1: length = 7 + 3 = 10; width = 7 – 2 = 5. **[3]**
21(a)	28 fish	M1: 20–25 part of the 15–25 bar (fd 3.6, width 5): 18. 25–30 part of the 25–40 bar (fd 2.0, width 5): 10. A1: total 18 + 10 = 28 fish. **[2]**
21(b)	15%	M1: fish under 15: bar 0-10 has fd 0.5, width 10: freq = 5. Bar 10-15 has fd 2.0, width 5: freq = 10. M1: total under 15 = 15 out of 100. A1: 15%. **[2]**
22(a)	Common ratio = 3	B1: 15/5 = 3 (or 45/15 = 3). **[1]**
22(b)	nth term = $5 \times 3^{n-1}$	M1: identifies first term 5 and ratio 3. A1: $5 \times 3^{n-1}$. **[2]**
22(c)	$k = 9$	M1: $5 \times 3^{k-1} = 32805$. M1: $3^{k-1} = 6561 = 3^8$. A1: $k - 1 = 8 \Rightarrow k = 9$. **[3]**

Worked Solutions — Predicted Paper 3 Set E

Higher Tier · Edexcel 1MA1 style

Q1: Prime factorisation

$168 = 2 \times 84 = 2 \times 2 \times 42 = 2 \times 2 \times 2 \times 21 = 2 \times 2 \times 2 \times 3 \times 7$.

$168 = 2^3 \times 3 \times 7$.

Q2: Inverse proportion

Total work = 6 machines × 14 hours = 84 machine-hours.

With 8 machines: time = 84 / 8 = **10.5 hours** (10 hours 30 minutes).

Q3: Linear inequality

$7 - 2x < 15 \Rightarrow -2x < 8$.

Divide by –2 (inequality reverses): **$x > -4$**.

Q4: Standard form

$(2.4 \times 5) \times (10^3 \times 10^4) = 12 \times 10^7$.

Adjust to standard form: **1.2×10^8**.

Q5: Two-way table probability

(a) P(pass) = 78/100 = **39/50** (or 0.78).

(b) Of 50 practical candidates, 8 failed. P(fail | practical) = 8/50 = **4/25**.

(c) P(pass) = 0.78; P(theory) = 50/100 = 0.50.

P(pass) × P(theory) = 0.78 × 0.50 = 0.39.

P(pass $\cap$ theory) = 36/100 = 0.36.

Since $0.36 \neq 0.39$, the events are **not independent**.

Q6: Percentage increase

New salary = 32500 × 1.04 = **£33,800**.

Q7: Quadratic factorising

Find two numbers that multiply to 21 and sum to –10: –3 and –7.

$(x - 3)(x - 7) = 0$.

$x = 3$ or $x = 7$.

Q8: Direct proportion (squared)

(a) $y = kx^2$. When $x = 4$, $y = 80$: $80 = 16k$, so $k = 5$.

$y = 5x^2$.

(b) $y = 5 \times 49 =$ **245**.

(c) $245 = 5x^2 \Rightarrow x^2 = 49 \Rightarrow$ **$x = 7$** (positive).

Q9: Quadratic graph

(a) Curve is below the x-axis between the roots –2 and 4.

$-2 < x < 4$.

(b) $x^2 - 2x - 8 = 7 \Rightarrow x^2 - 2x - 15 = 0$.

Factorise: $(x - 5)(x + 3) = 0$.

$x = 5$ or $x = -3$.

Q10: Rationalising

$12/\sqrt{6} \times \sqrt{6}/\sqrt{6} = (12\sqrt{6}) / 6 = \mathbf{2\sqrt{6}}$.

Q11: Speed-distance equation

Let d = distance one way (in km).

Time out = d/75 hours; time back = d/60 hours.

$d/75 + d/60 = 4.5$.

Common denominator 300: $(4d)/300 + (5d)/300 = 4.5 \Rightarrow 9d/300 = 4.5$.

$9d = 1350 \Rightarrow$ **d = 150 km**.

(Check: time out = 150/75 = 2 h. Time back = 150/60 = 2.5 h. Total = 4.5 h. ✓)

Q12: Probability tree

(a) P(cycle) = P(rain) × P(cycle | rain) + P(no rain) × P(cycle | no rain).

= 0.3 × 0.4 + 0.7 × 0.6 = 0.12 + 0.42 = **0.54**.

(b) P(rain | cycle) = P(rain ∩ cycle) / P(cycle) = 0.12 / 0.54 = **2/9** ≈ **0.222** (3 s.f.).

Q13: Algebraic fractions

Numerator (difference of squares): $4x^2 - 1 = (2x - 1)(2x + 1)$.

Denominator: $2x^2 + 5x - 3$. Factors of $2 \times (-3) = -6$ summing to 5: +6 and –1.

Split: $2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3)$.

Cancel $(2x - 1)$: **$(2x + 1) / (x + 3)$**.

Q14: Vectors in a rectangle

(a) In rectangle OABC, B = **a** + **c**.

M is midpoint of AB: **OM** = (1/2)(**OA** + **OB**) = (1/2)(**a** + **a** + **c**) = **a** + (1/2)**c**.

N is midpoint of BC: **ON** = (1/2)(**OB** + **OC**) = (1/2)(**a** + **c** + **c**) = (1/2)**a** + **c**.

MN = **ON** – **OM** = (1/2)**a** + **c** – **a** – (1/2)**c** = –**(1/2)a + (1/2)c**.

(b) CA = **OA** – **OC** = **a** – **c**.

MN = (1/2)(–**a** + **c**) = –(1/2)(**a** – **c**) = –(1/2)**CA**.

So **MN is parallel to CA** (one of the diagonals of the rectangle) and has half its length.

Q15: Rationalising

Multiply top and bottom by conjugate $(3 + \sqrt{5})$:

$4(3 + \sqrt{5}) / ((3 - \sqrt{5})(3 + \sqrt{5})) = (12 + 4\sqrt{5}) / (9 - 5) = (12 + 4\sqrt{5}) / 4$.

Divide each term by 4: **$3 + \sqrt{5}$** (so a = 3, b = 1).

Q16: Line through midpoint

(a) Gradient L = (–1 – 3) / (8 – 2) = –4/6 = **–2/3**.

(b) Midpoint of AB = ((2+8)/2, (3+(–1))/2) = (5, 1).

Perpendicular gradient = 3/2 (negative reciprocal).

y – 1 = (3/2)(x – 5).

y = (3/2)x – 15/2 + 1 = **(3/2)x** – **13/2** (or 1.5x – 6.5).

Q17: Sine equation

(a) $\sin^{-1}(0.5) = 30°$ (principal value).

Second solution in range: 180 – 30 = 150°.

x = 30° or x = 150°.

(b) sin is negative in 3rd and 4th quadrants; reference angle 30°.

x = 210° or x = 330°.

Q18: Compound percentage change

(a) Combined multiplier = 1.05 × 1.08 × 1.06 = 1.20204.

Percentage increase = (1.20204 – 1) × 100 = **20.20%** (2 d.p.).

(b) Value after 3 years = 4000 × 1.20204 = **£4,808.16**.

Q19: Two tangents from external point

(a) In right triangle OAP (right angle at A, since tangent ⊥ radius):

sin(angle OPA) = OA / OP = 2.5 / 5 = 0.5.

angle OPA = $\sin^{-1}(0.5)$ = **30°**.

(b) By symmetry, the two tangents make equal angles with OP. **Angle APB = 2 × 30 = 60°**.

(c) $PA^2 = OP^2 – OA^2$ = 25 – 6.25 = 18.75.

PA = √18.75 = √(75/4) = (√75)/2 = **(5√3)/2 cm** ≈ 4.33 cm.

Q20: Quadratic word problem

(a) Area = length × width: (x + 3)(x – 2) = 50.

Expand: $x^2 + x - 6 = 50 \Rightarrow$ **$x^2 + x - 56 = 0$**. ✓

(b) Factorise: (x + 8)(x – 7) = 0. So x = 7 or x = –8.

Reject x = –8 (lengths must be positive). x = 7.

Length = 10 cm, width = 5 cm.

(Check: 10 × 5 = 50. ✓)

Q21: Histogram

Frequency = frequency density × class width:

0–10: 0.5 × 10 = 5; 10–15: 2.0 × 5 = 10; 15–25: 3.6 × 10 = 36;

25–40: 2.0 × 15 = 30; 40–60: 0.8 × 20 = 16.

Total = 97 (close to 100; minor rounding).

(a) 20–30 spans half of the 15–25 bar (3.6 × 5 = 18) + part of the 25–40 bar (2.0 × 5 = 10).

Estimate ≈ 18 + 10 = **28 fish**.

(b) Fish shorter than 15: 5 (from 0–10) + 10 (from 10–15) = 15.

15% of all fish.

Q22: Geometric sequence

(a) Common ratio = 15/5 = **3** (or 45/15 = 3).

(b) First term a = 5, ratio r = 3. nth term = ar^{n-1} = $\mathbf{5 \times 3^{n-1}}$.

(c) $5 \times 3^{k-1} = 32805$.

$3^{k-1} = 32805 / 5 = 6561 = 3^8$.

$k - 1 = 8 \Rightarrow$ $\mathbf{k = 9}$.